

Arnav Goyal

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TECHNICAL SKILLS

Languages: Python, C/C++, SQL, PL/SQL, R, MATLAB

Frameworks & Libraries: FastAPI, TensorFlow, Keras, PyTorch, Ultralytics, ByteTrack, Supervision, LlamaIndex

Developer Tools: Git, VS Code, PyCharm, Jupyter Notebook, Google Colab, Docker

Databases & Storage: PostgreSQL, MongoDB, ChromaDB, SQLite

Concepts: Data Structures, OOPS, DBMS, Operating Systems, Deep Learning, Machine Learning, REST APIs

EDUCATION

Thapar University

Bachelor of Engineering in Computer Engineering — CGPA: 8.59

Patiala, Punjab

Expected June 2027

WORK EXPERIENCE

AI/ML Intern

June 2025 – Present

OneAssure, Bengaluru

Remote

- Optimized a **RAG** pipeline for policy extraction using **LlamaIndex**, **OpenAI**, and **Gemini** fallback.
- Patched core modules to reduce field-level retrieval errors by 35% across varied insurance formats.
- Constructed a scalable user authentication system using **FastAPI** and **MongoDB** which is now used by over 20 internal applications and protects sensitive user data.
- Guided **JSON** evaluation suite achieving 91%+ field-wise match rate on unseen documents.

UG Research Assistant

Jan. 2025 – May 2025

Under: Dr. Ravneet Kaur

Thapar University, Patiala

- Innovated a custom lightweight hybrid **CNN model** with only **1.6M parameters** for **real-time** trajectory prediction reaching **240+ FPS**.
- Achieved **94.2% mAP@50** on **KITTI** with a highly optimized detection pipeline.
- Designed **LSTM-based** trajectory predictor reaching 0.66 ADE **over 5 future steps**.
- Integrated **TensorRT** and reduced Kalman filter RMSE to 2.21 for smooth tracking.

PROJECTS

EcoCart – Smart Retail & Sustainability Advisor | *YOLOv8, FastAPI, OpenAI, ChromaDB*

July 2025

- Developed a vision module using **YOLOv8** and **ByteTrack** for real-time product detection in smart carts.
- Built a **FastAPI** backend with **OpenAI-powered LLM** agents for advising eco-friendly alternatives.
- Integrated **ChromaDB vector store** for sustainability metadata retrieval and **RAG-based** recommendations.
- Achieved 92%+ SKU detection accuracy and near-instant LLM response latency (<1s) in live tests.

ADR Prediction using Graph Neural Networks | *PyTorch Geometric, GNN, RAG*

Apr 2025 – May 2025

- Built a **GNN-based model** on a biomedical graph with 20K drugs to predict adverse drug reactions.
- Achieved **99.1% AUC** and **98.5% AUPRC**, outperforming baselines by **15%**.
- Integrated **RAG** for explainability, improving rare ADR detection by 18% and Reducing false positives by 30%.

Real-Time Exam Cheating Detection | *Python, OpenCV, YOLOv8, CLI Tool*

Jan 2025 – Apr 2025

- Developed a **real-time CLI** tool with 92% accuracy using **YOLOv8** and **MediaPipe**.
- Detected 5+ unauthorized objects in live video streams with high precision reaching **50 fps**.
- Trained on **5,000+ frames**, achieving an **F1-score of 0.88**.

ACHIEVEMENTS

- Solved 300+ algorithmic problems, translating complex challenges into efficient, scalable solutions — ready for real-world software development. <https://leetcode.com/u/arnavg566/>
- Won the Kaggle Hack organized by Thapar Institute, outperforming 100+ participants with a top-ranking solution.